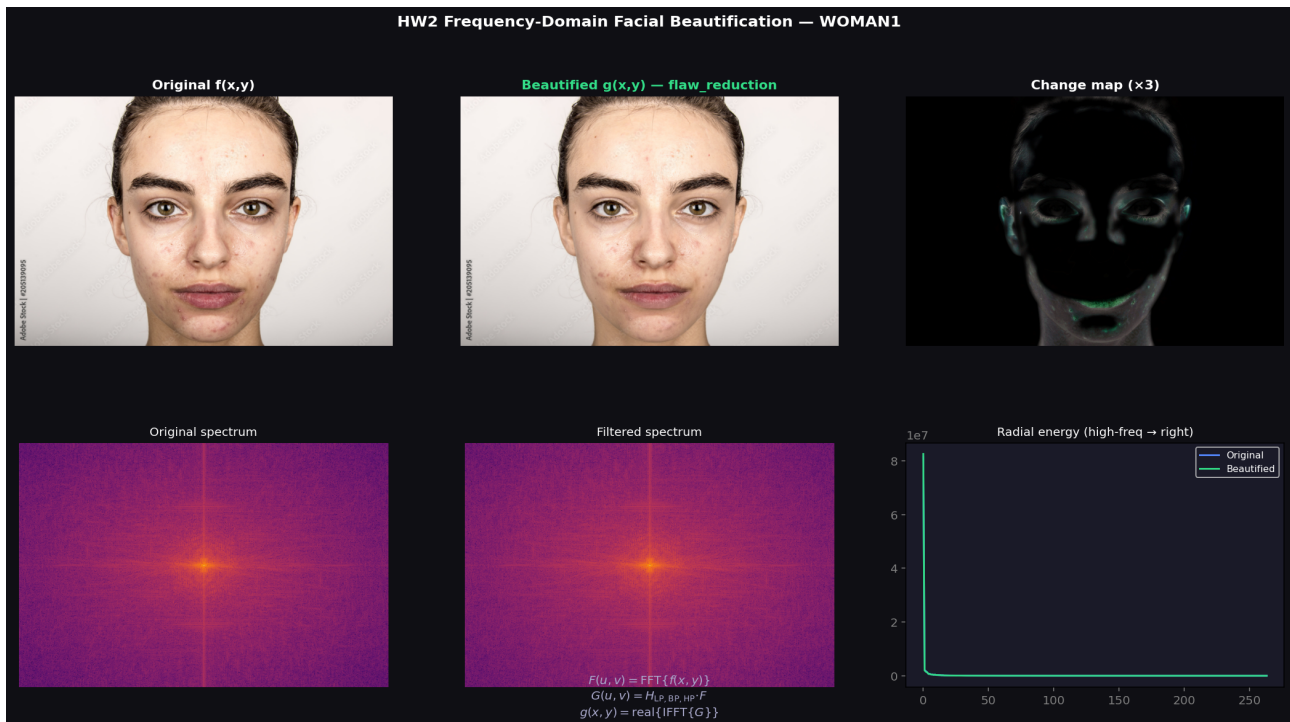


Homework 2: Frequency-Domain Facial Beautification

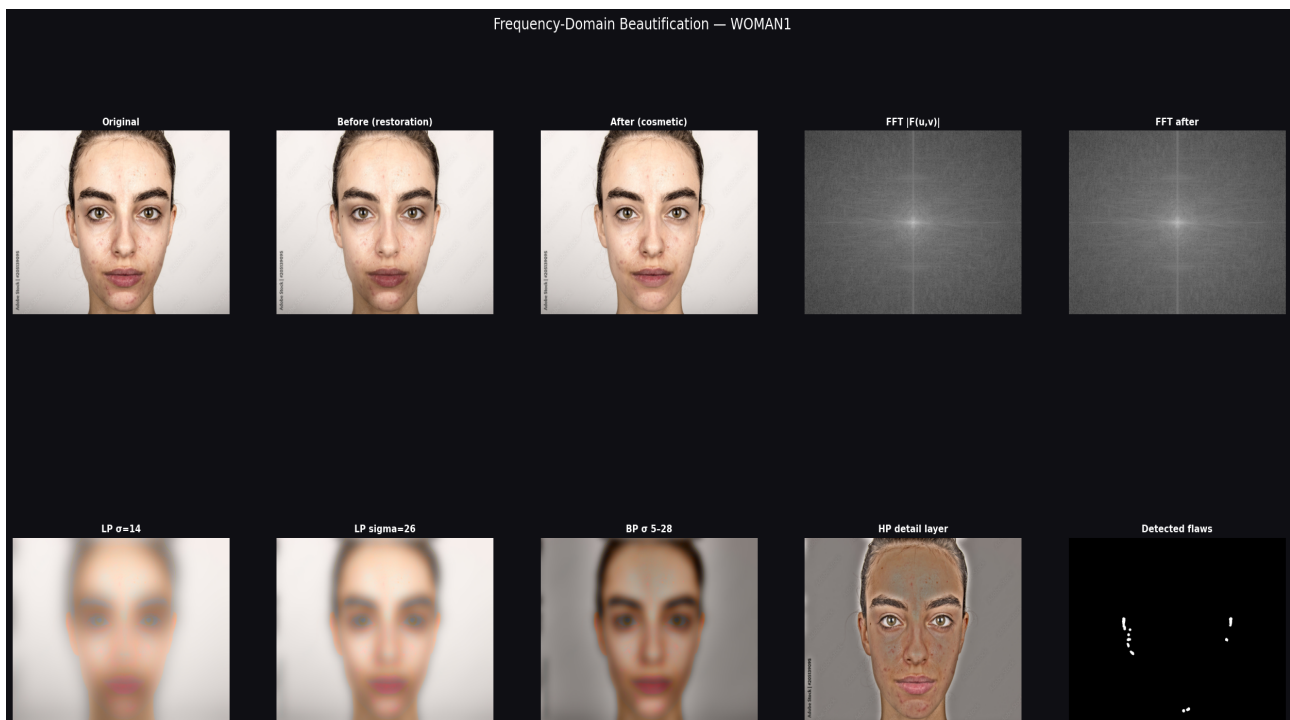
Pipeline: universal flaw_reduction — auto face analysis, neon-box before preview, adaptive line-flaw smoothing (bbox-relative), gradient LAB fix for red blemishes.

$$F(u,v)=\text{FFT}\{f\}; \quad G=H \cdot F; \quad g=\text{real}\{\text{IFFT}\{G\}\}$$

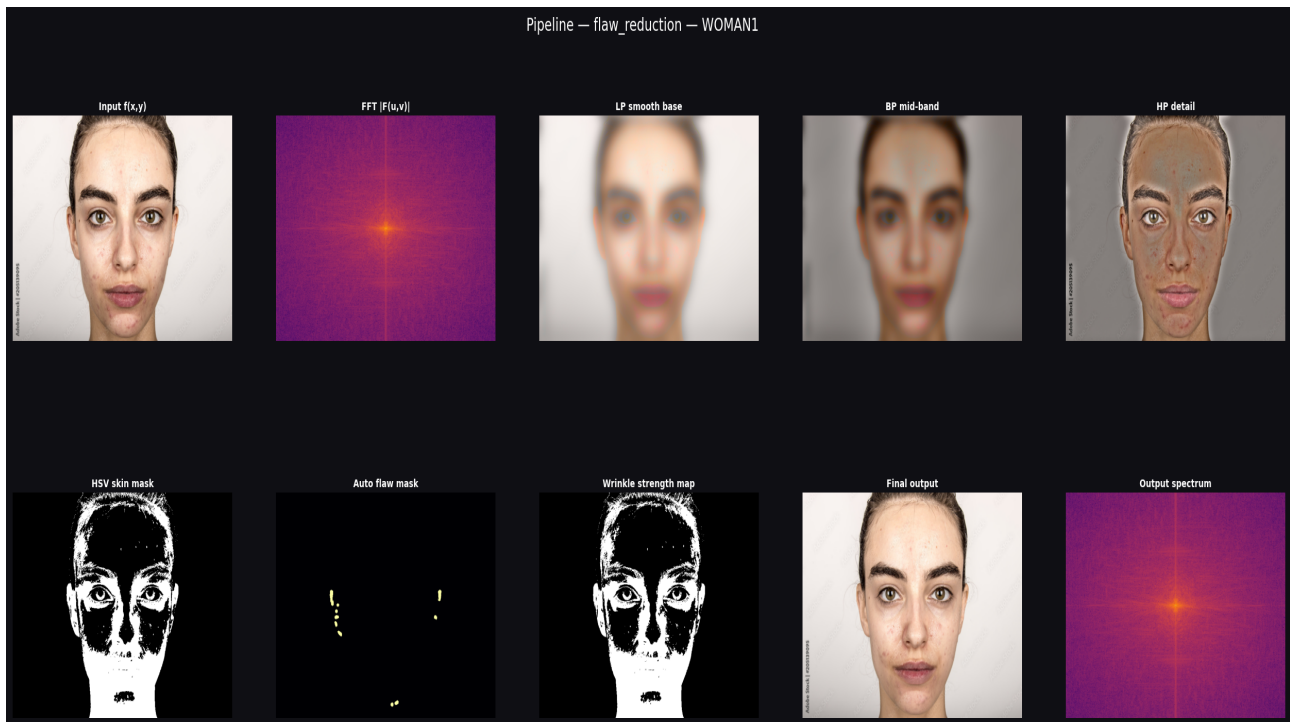
Results — WOMAN1 (flaw_reduction)



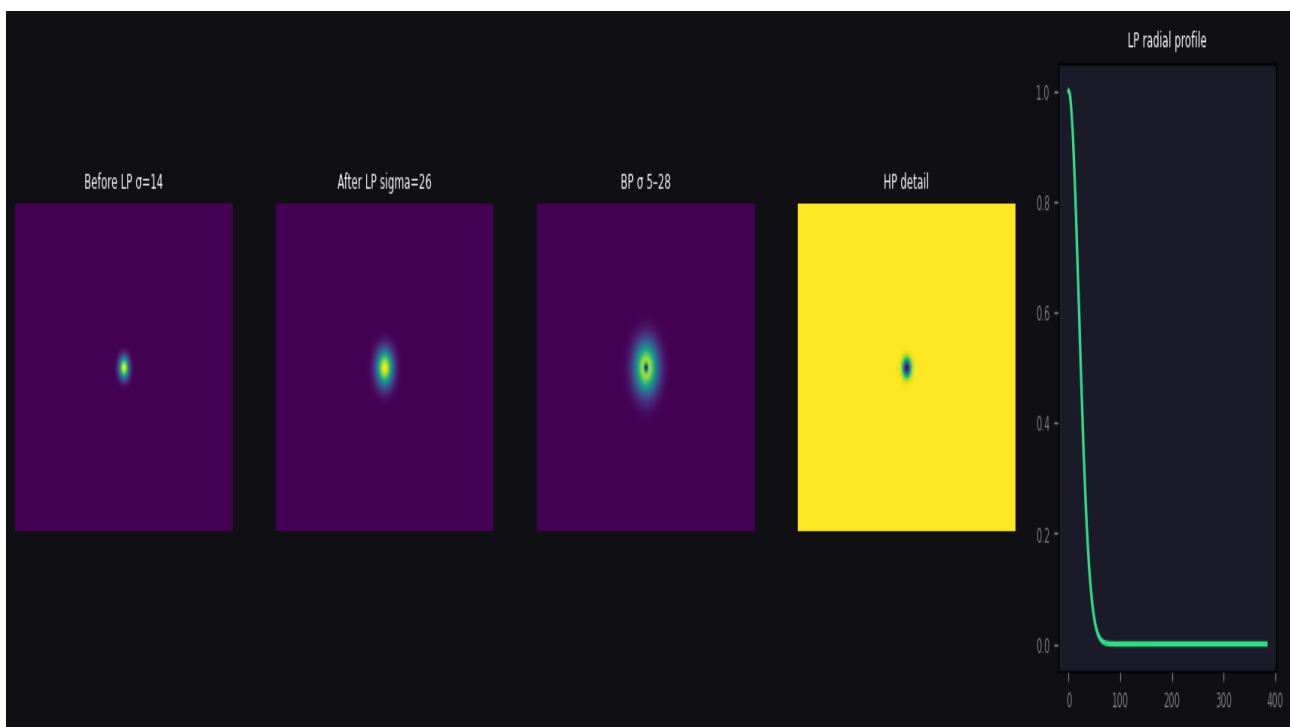
Professor summary: original vs beautified, FFT equation, radial energy.



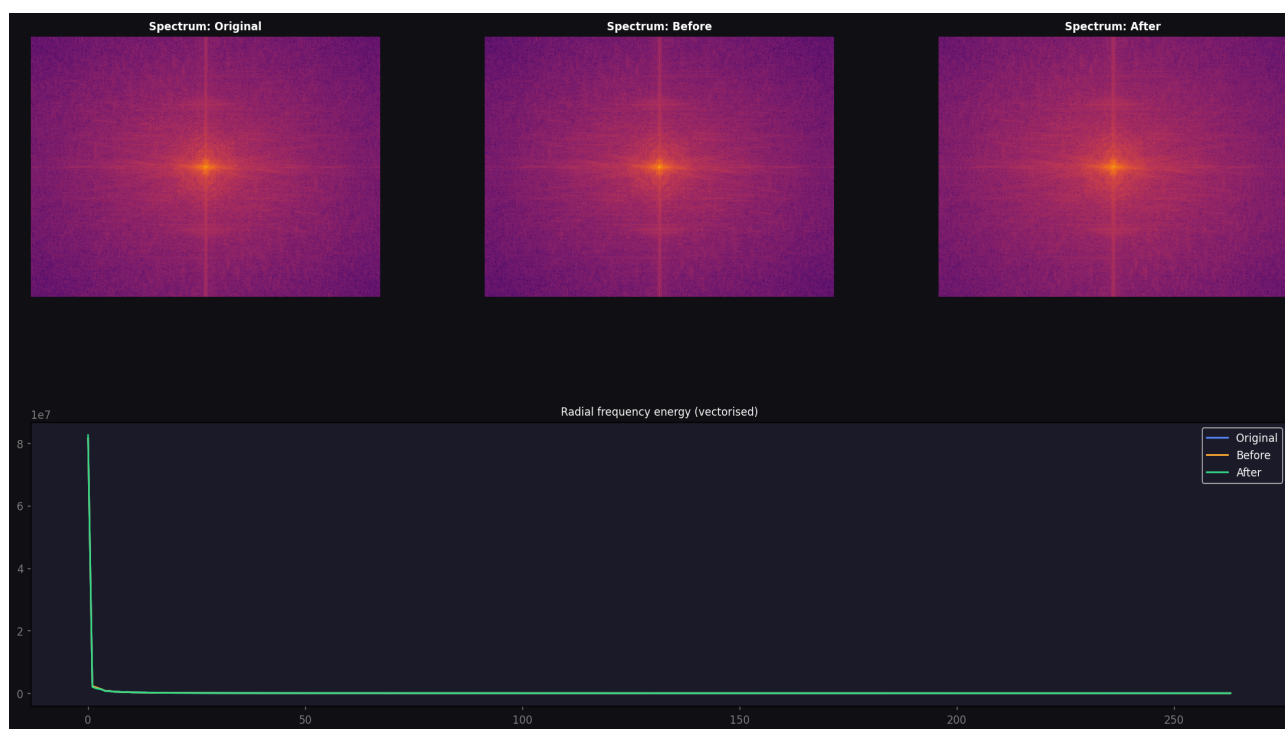
Full comparison with adaptive weight map.



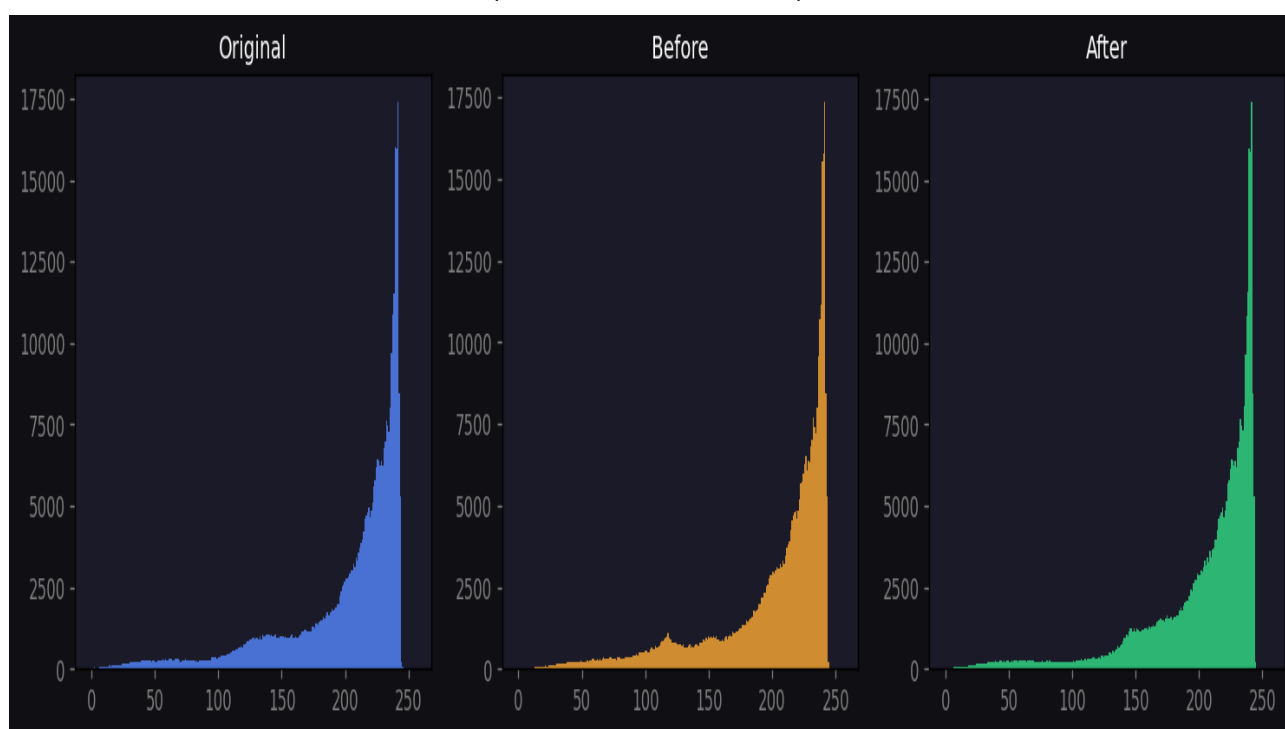
Pipeline steps and masks.



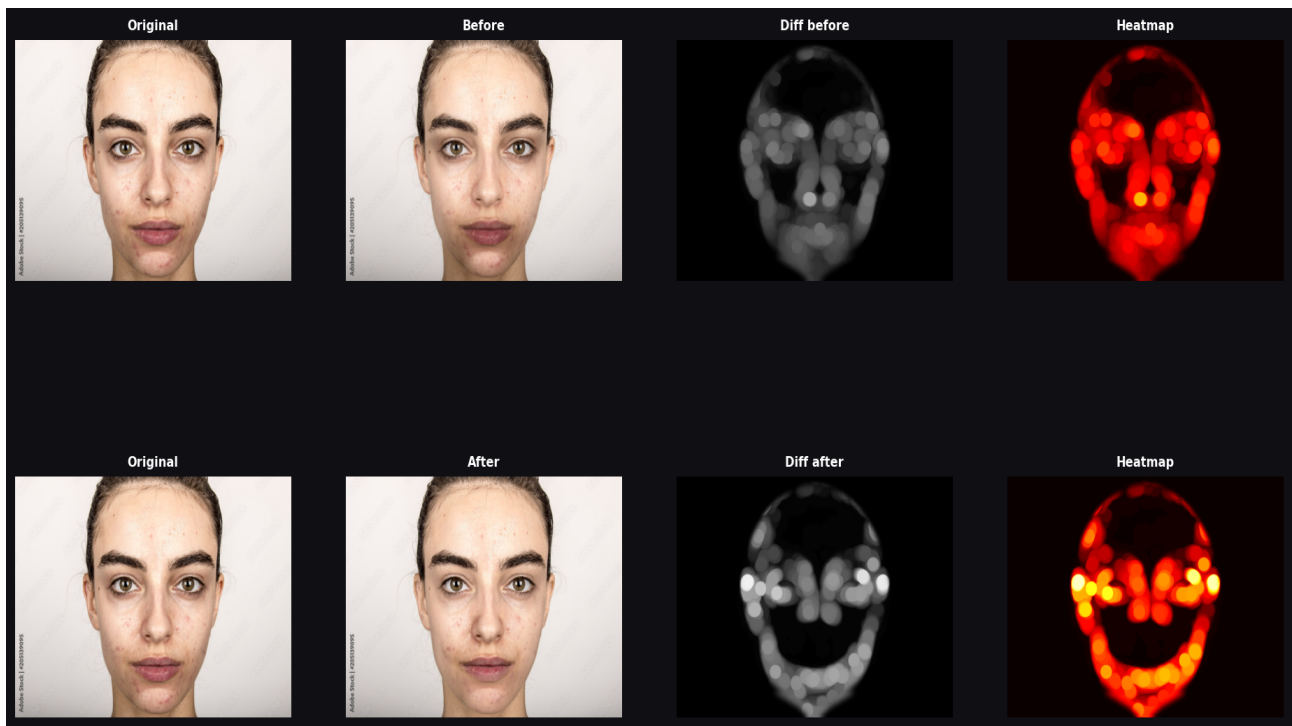
Filter kernels $H(u,v)$.



FFT spectra and vectorised radial profile.



Histograms.



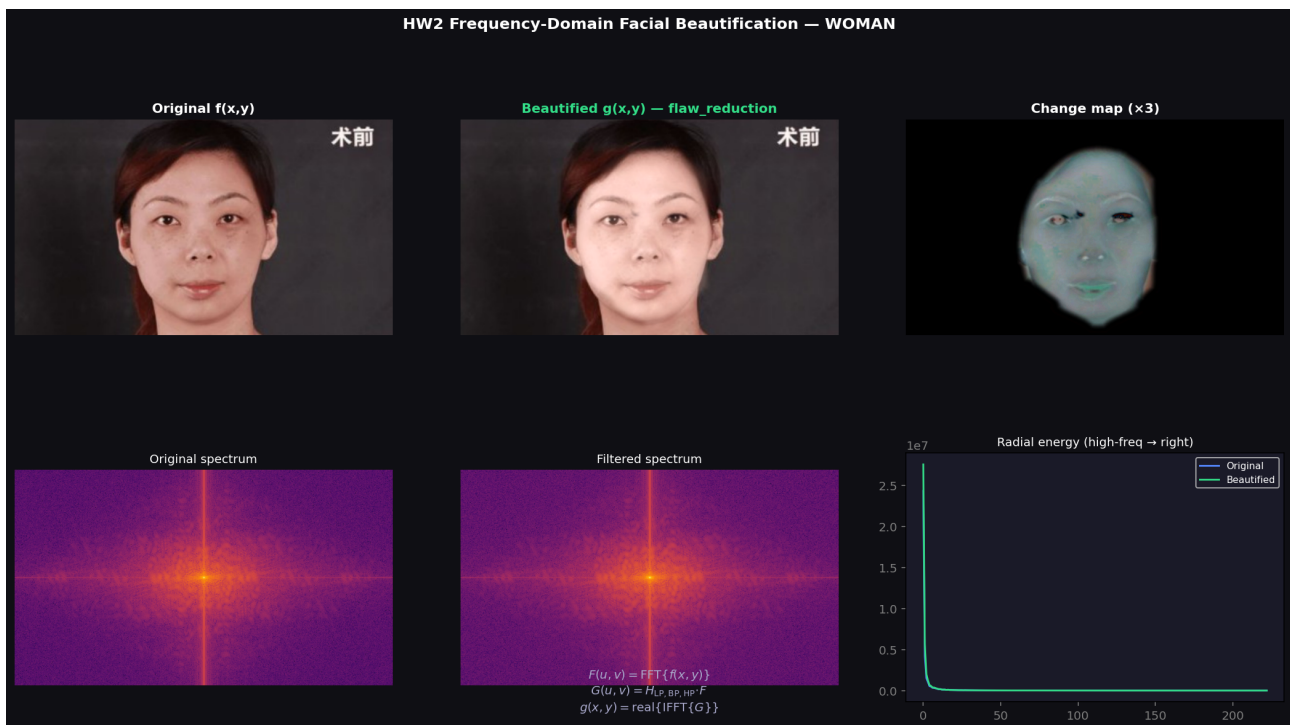
Difference maps.



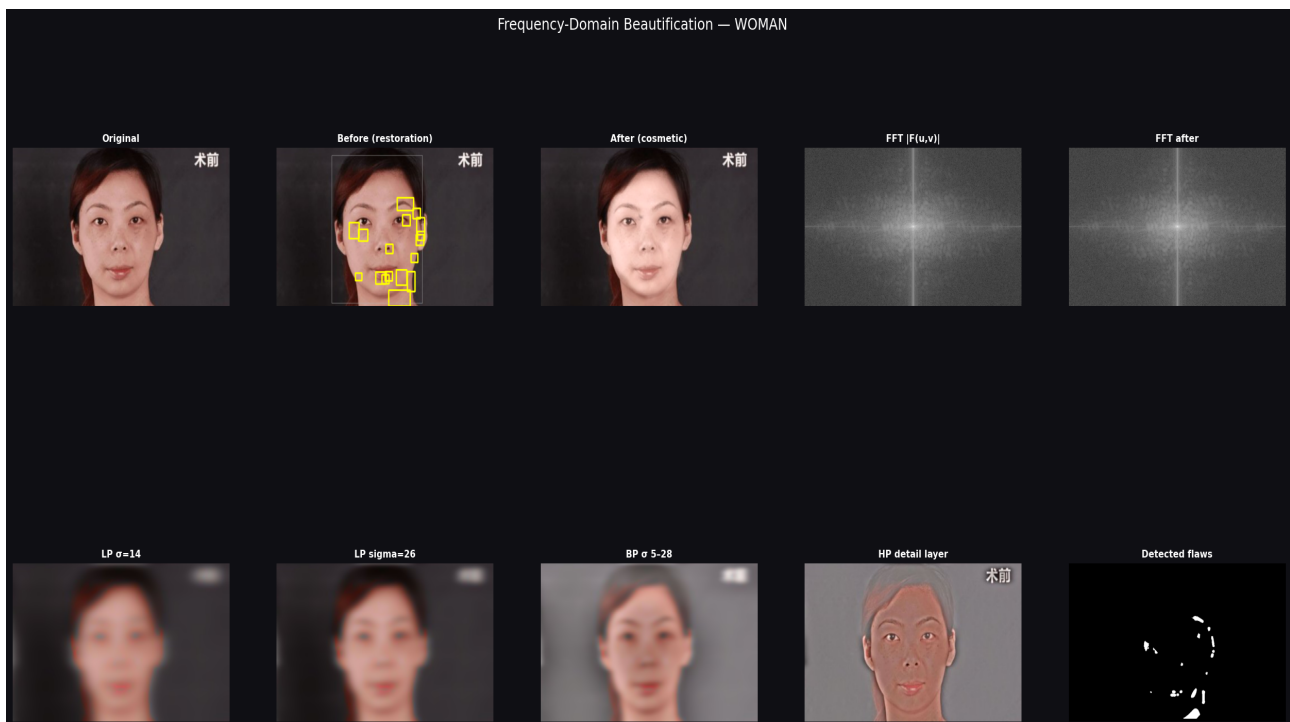
RGB channel analysis.

Metric	Before	After
PSNR	33.55	31.65
SSIM	0.9918	0.9901
Std-dev	51.16	48.60

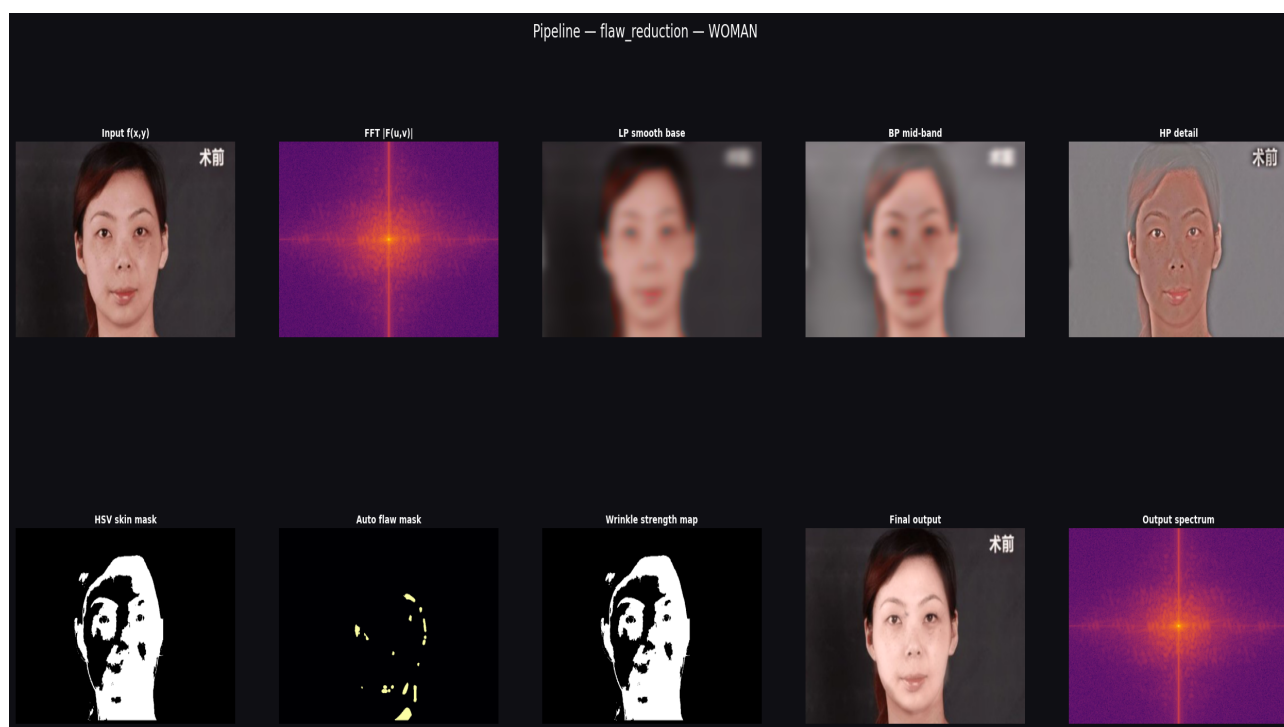
Results — WOMAN (flaw_reduction)



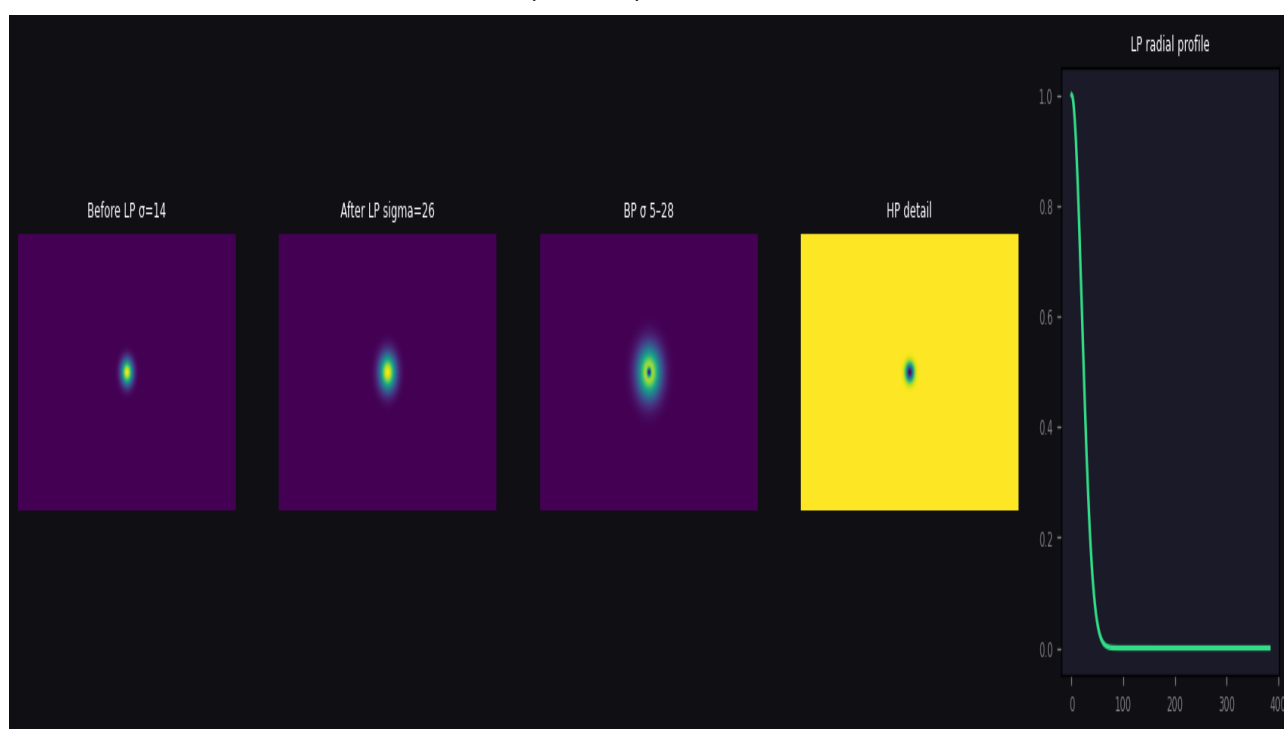
Professor summary: original vs beautified, FFT equation, radial energy.



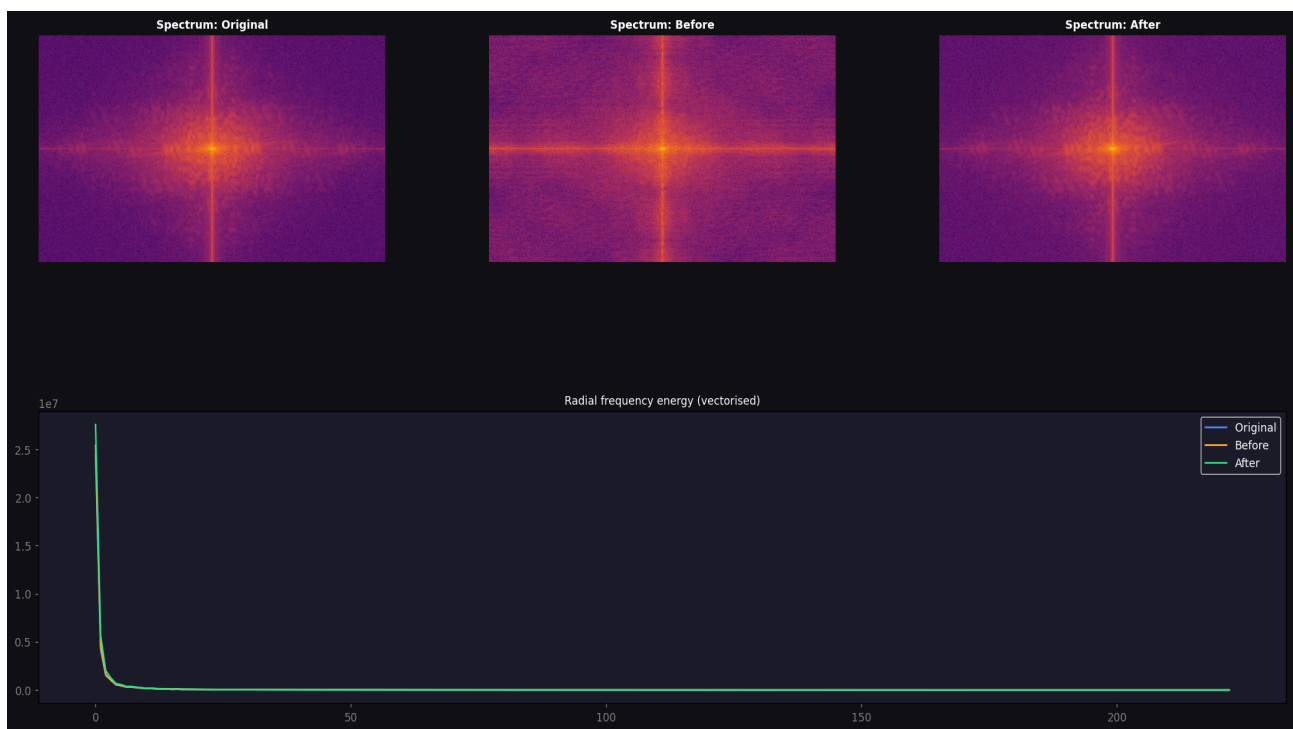
Full comparison with adaptive weight map.



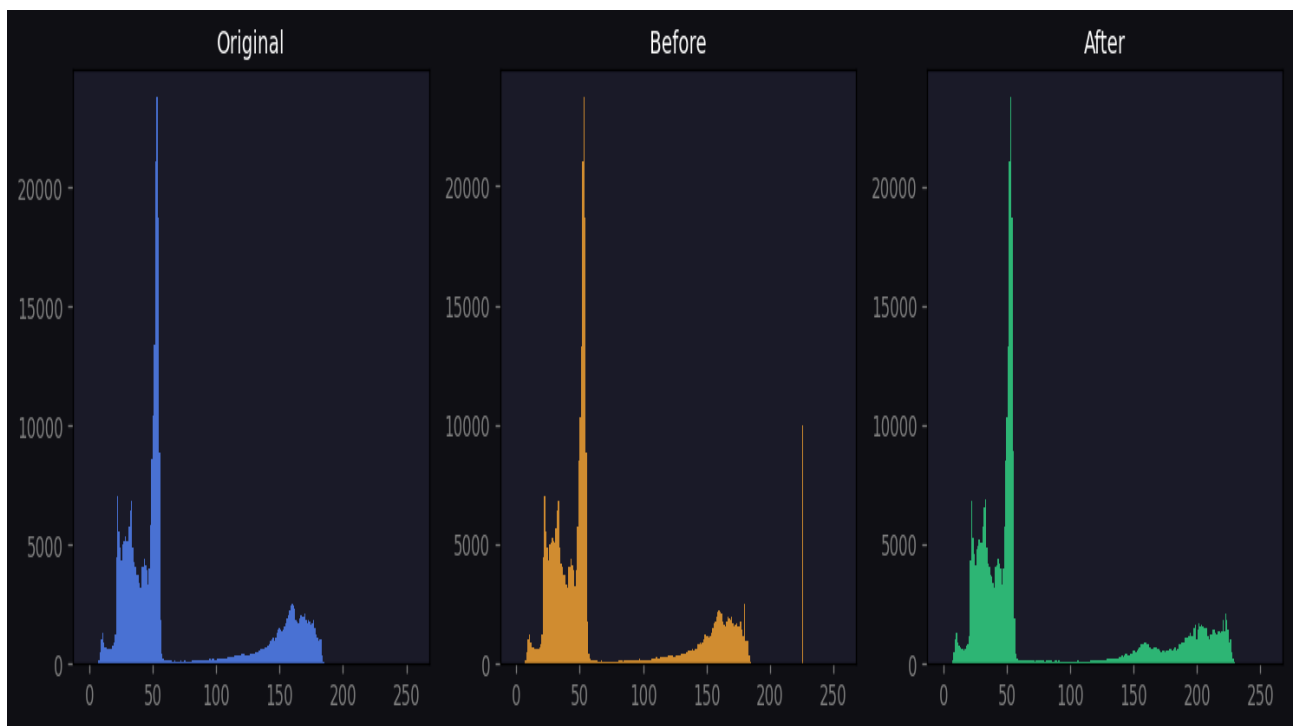
Pipeline steps and masks.



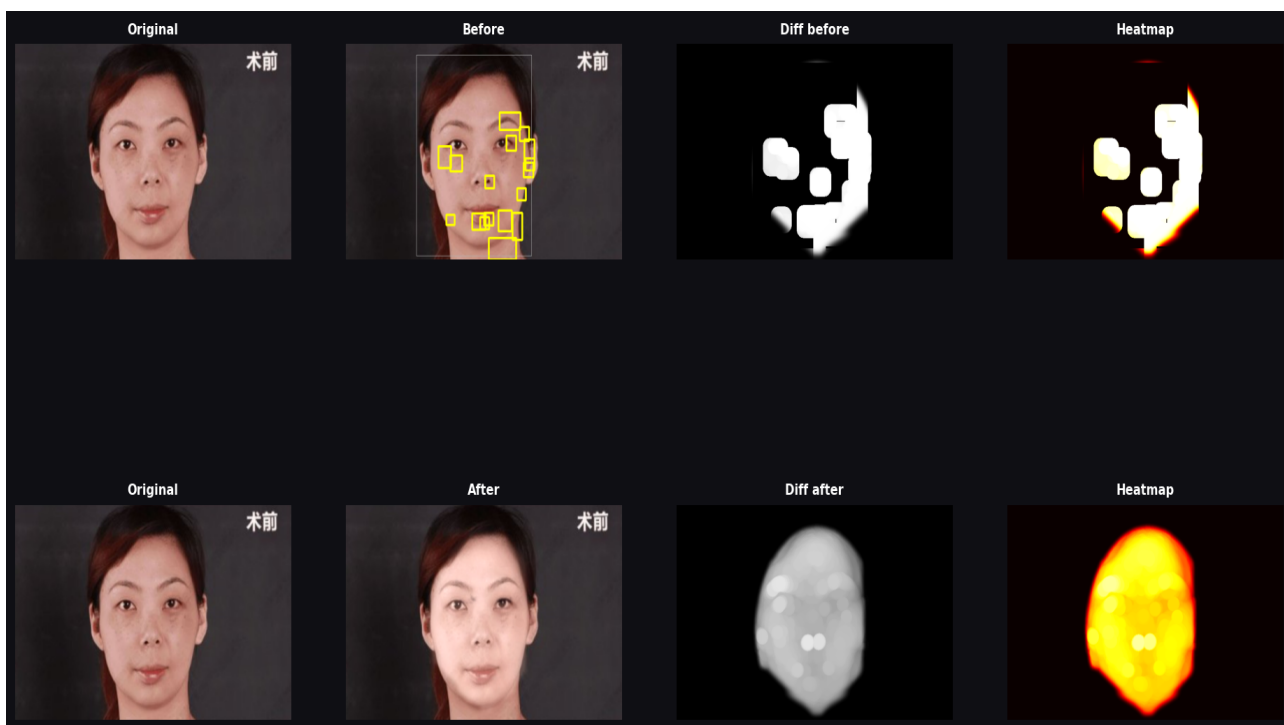
Filter kernels $H(u,v)$.



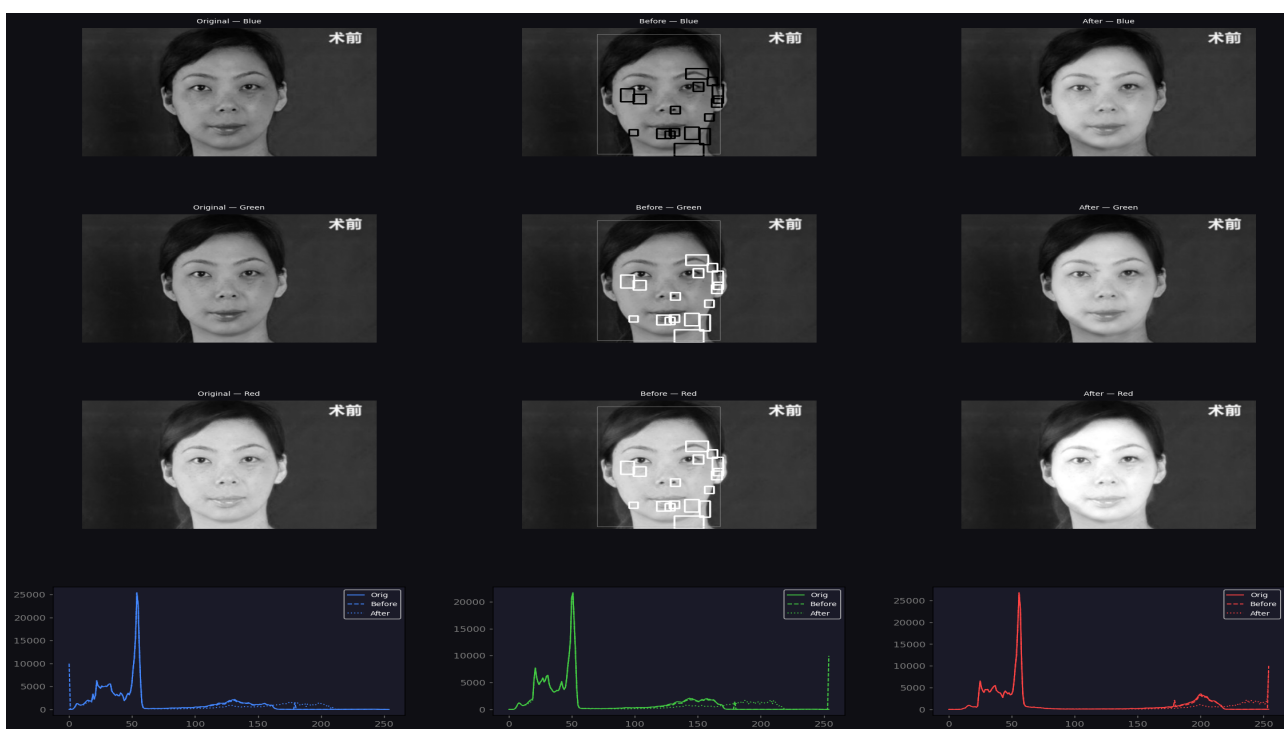
FFT spectra and vectorised radial profile.



Histograms.



Difference maps.



RGB channel analysis.

Metric	Before	After
PSNR	21.54	22.28
SSIM	0.9177	0.9808
Std-dev	59.05	68.40